

Nonlinear Classification of Circular Data Using Support Vector Machines

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Abstract—This paper presents a study of Support Vector Machine (SVM) classifiers applied to a synthetic two-class circular dataset that is not linearly separable in 2D. We evaluate a linear kernel, polynomial kernels across four degrees (2, 5, 7, 10), and RBF kernels across three gamma values (10, 30, 100), reporting both training and testing misclassification errors. Results show that even-degree polynomial kernels naturally match the circular symmetry of the data and achieve zero error, while odd-degree kernels consistently underfit. For RBF kernels, moderate gamma values generalize well, but a high gamma of 100 leads to overfitting with a testing error of 16.67%. The linear kernel fails entirely on this problem, confirming that the data requires a nonlinear decision boundary.

Index Terms—Support Vector Machine, kernel methods, polynomial kernel, RBF kernel, nonlinear classification, overfitting, misclassification error

I. INTRODUCTION

In many real-world classification problems, the data cannot be separated using a straight line or a flat surface. When this happens, a simple linear classifier will always fail, no matter how well it is trained. Support Vector Machines (SVMs) handle this by using what is called the kernel trick, which maps the data into a higher-dimensional space where a linear boundary can do the job [1].

The choice of kernel and its settings matters a lot. If the kernel is too simple, the model will underfit and make too many mistakes on both training and testing data. If the kernel is too complex, it will memorize the training data and fail on new, unseen points — this is called overfitting [2]. Finding the right balance between these two extremes is one of the main challenges in applying SVMs in practice [3].

In this paper, we explore these ideas using a synthetic dataset made up of two concentric circles. This type of data is a classic example of a problem that is not linearly separable in 2D. We test a linear kernel, polynomial kernels with four different degrees, and RBF kernels with three different gamma values, and we compare how each one performs in terms of misclassification error and decision boundary shape. Overfitting occurs when a model becomes excessively complex and begins to capture noise in the training data, resulting in poor generalization to unseen samples.

II. DATA CREATION

In this experiment, we created a dataset with two groups of points forming circles: an inner circle and an outer circle.

To generate points on a circle, we used angles and radii. First, we randomly picked angles between 0 and 2π . Then, we converted those angles into coordinates using sine and cosine:

$$x_1 = r \cos(\theta), \quad x_2 = r \sin(\theta)$$

This gives us points around a circle, where x_1 is the horizontal position and x_2 is the vertical position. We also added a small amount of noise to the radius so the data looks more realistic and not perfectly uniform.

For the inner circle, we used a smaller radius (around 1), so the points stay close to the center. For the outer circle, we used a larger radius (around 3), creating a bigger circle around it. Finally, we combined both sets of points into one dataset and assigned labels:

- Inner circle $\rightarrow$ Class 0
- Outer circle $\rightarrow$ Class 1

This creates a circular dataset where the two classes cannot be separated using a straight line in 2D.

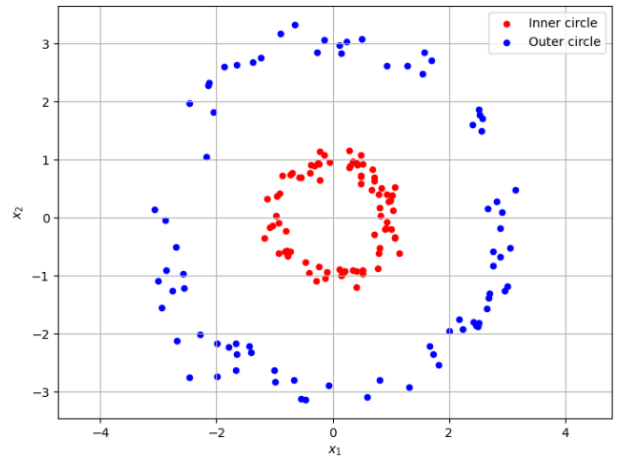


Fig. 1. Two-dimensional circular dataset consisting of an inner and outer circle. The classes are not linearly separable in this space.

Figure 1 shows the original dataset in two dimensions. It is clear that the inner and outer circles cannot be separated using a straight line, confirming that a nonlinear approach is needed.

III. LINEAR KERNEL SVM

Before testing nonlinear kernels, we first applied a linear SVM directly to the 2D data to establish a baseline. As expected, the linear kernel completely fails on this problem. The decision boundary it produces is a straight line — a roughly horizontal cut across the feature space — which cannot wrap around the inner circle in any direction.

TABLE I
LINEAR KERNEL SVM PERFORMANCE

Kernel	Train Miscls	Test Miscls	Train Error	Test Error
Linear	32	16	0.2857	0.3333

Table I shows that the linear SVM produces high misclassification errors on both training and testing data.

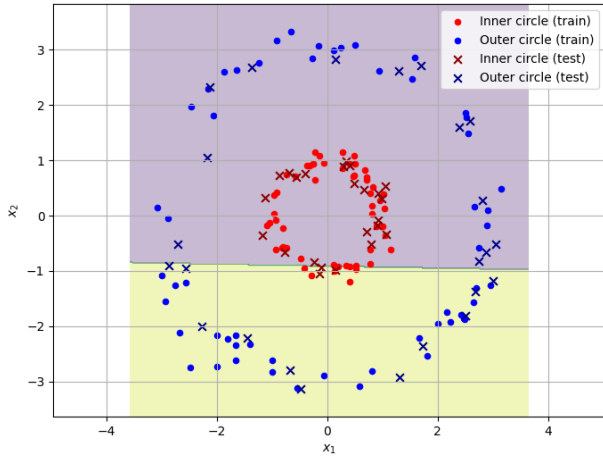


Fig. 2. Linear Kernel SVM decision boundary on the circular dataset. The straight-line boundary fails to separate the two classes, resulting in high training and testing errors.

Figure 2 shows the decision boundary produced by the linear SVM. The purple and yellow regions represent the two predicted classes. A large portion of both the inner and outer circle points fall on the wrong side of the boundary.

The training error is approximately 28.57%, while the testing error is approximately 33.33%, which are among the highest across all models tested. This result is expected because the circular structure of the data cannot be represented by a linear boundary. As a result, the model underfits and fails to capture the underlying pattern of the data [4].

IV. POLYNOMIAL KERNEL RESULTS

A. Summary of Results

TABLE II
POLYNOMIAL KERNEL SVM PERFORMANCE FOR DIFFERENT DEGREES

Degree	Train Miscls	Test Miscls	Train Error	Test Error
2	0	0	0.0000	0.0000
5	26	14	0.2321	0.2917
7	26	12	0.2321	0.2500
10	0	1	0.0000	0.0208

Table II summarizes the performance of the polynomial kernel SVM for different degrees. Even degrees (2, 10) perform significantly better than odd degrees (5, 7), which struggle to capture the circular structure of the data.

The better performance of even-degree polynomials can be explained by the symmetry of the data. Since the dataset forms concentric circles, it is naturally symmetric around the origin. Even-degree terms such as x_1^2 and x_2^2 preserve this symmetry because they always produce non-negative values and depend only on the magnitude of the input, resulting in circular decision boundaries [3].

In contrast, odd-degree terms such as x_1^3 or x_2^3 change sign depending on whether the input is positive or negative. This sign change introduces asymmetry in the model, which distorts the decision boundary and prevents it from properly capturing the circular structure of the data. Therefore, even-degree polynomial kernels are more suitable for this problem.

B. Degree = 2

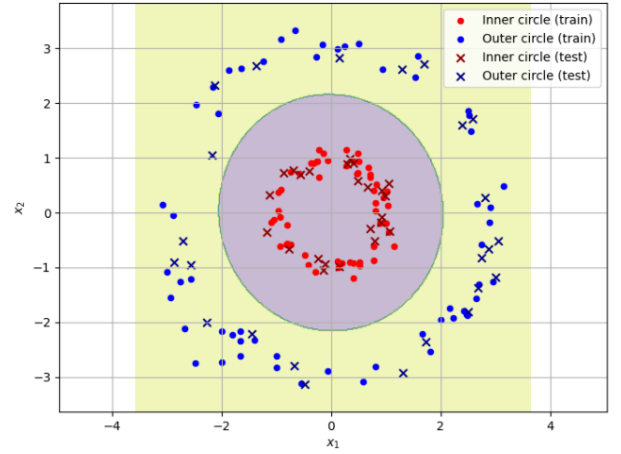


Fig. 3. Polynomial Kernel SVM with degree = 2.

The decision boundary forms a smooth circular shape that perfectly separates the inner and outer circles. Both training and testing errors are zero, indicating an excellent fit. This confirms that a low-degree polynomial is sufficient when its structure aligns with the symmetry of the data.

C. Degree = 5

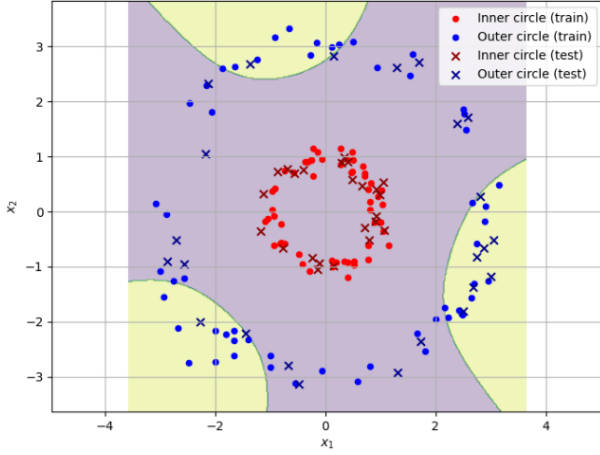


Fig. 4. Polynomial Kernel SVM with degree = 5.

The boundary becomes irregular and fails to match the circular structure of the data. Both training and testing errors are high, indicating underfitting. The failure here is not due to insufficient complexity, but rather a mismatch between the kernel structure and the data geometry.

D. Degree = 7

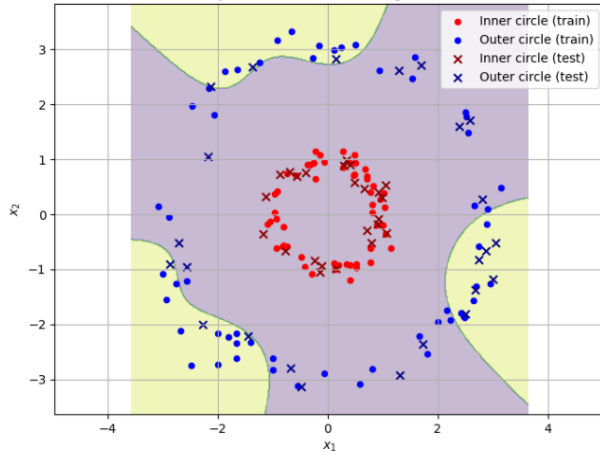


Fig. 5. Polynomial Kernel SVM with degree = 7.

The boundary is distorted and asymmetric, leading to higher training and testing errors. This indicates underfitting. The asymmetric behavior introduced by odd-degree terms leads to distorted boundaries that fail to capture the circular pattern.

E. Degree = 10

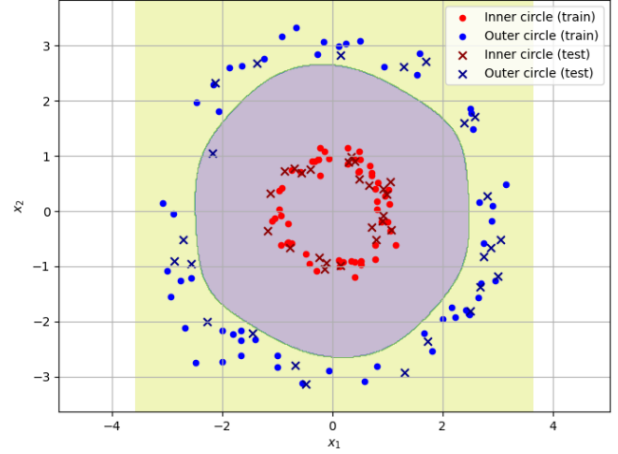


Fig. 6. Polynomial Kernel SVM with degree = 10.

The model produces a circular boundary but with slightly increased complexity. While training error is zero, a small testing error appears, indicating the beginning of overfitting.

V. RBF KERNEL RESULTS

A. Summary of Results

TABLE III
RBF KERNEL SVM PERFORMANCE FOR DIFFERENT GAMMA VALUES

Gamma	Train Miscls	Test Miscls	Train Error	Test Error
10	0	0	0.0000	0.0000
30	0	0	0.0000	0.0000
100	0	8	0.0000	0.1667

Table III summarizes the performance of the RBF kernel SVM for different values of γ . Moderate values of γ (10 and 30) achieve perfect classification on both training and testing data, indicating a good fit. As γ increases to 100, the model becomes overly sensitive to individual data points and the testing error rises significantly [2].

B. $\Gamma = 10$

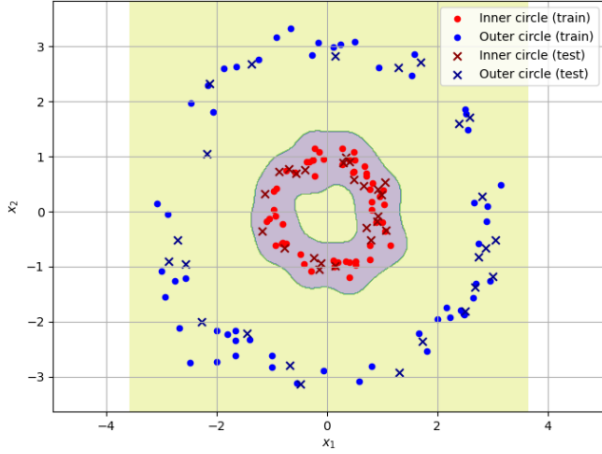


Fig. 7. RBF Kernel SVM with $\gamma = 10$.

For $\gamma = 10$, the decision boundary forms a smooth circular shape that correctly separates the inner and outer circles. The model generalizes well, achieving zero training and testing errors. This reflects a good balance between flexibility and smoothness, allowing the model to capture the global structure of the data without becoming overly sensitive to local variations.

C. $\Gamma = 30$

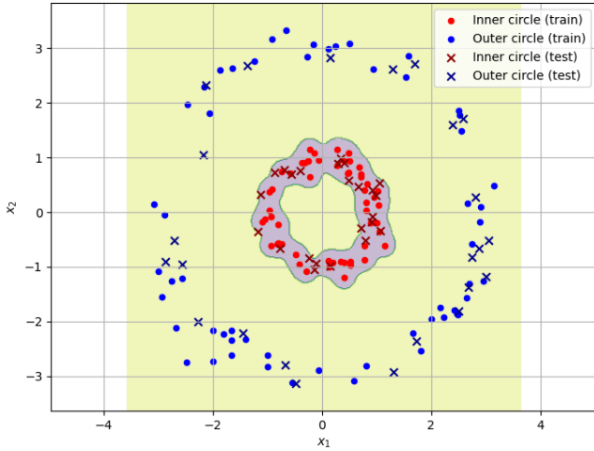


Fig. 8. RBF Kernel SVM with $\gamma = 30$.

For $\gamma = 30$, the boundary becomes slightly more flexible and closely follows the shape of the data. The model still maintains perfect performance, indicating that it is capturing the structure of the data effectively without overfitting. This demonstrates that increased flexibility can still maintain generalization when the model does not become overly localized.

D. $\Gamma = 100$

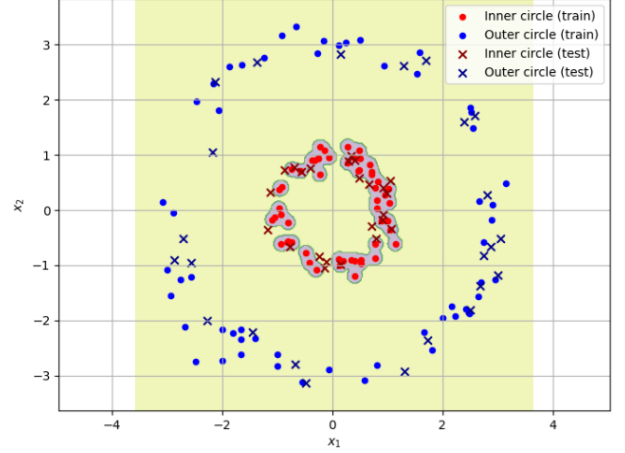


Fig. 9. RBF Kernel SVM with $\gamma = 100$.

As shown in Figure 9, the decision boundary becomes highly irregular and tightly wraps around individual data points. While the training error remains zero, the testing error rises to 16.67%, indicating clear overfitting.

The model becomes too sensitive to small variations and noise in the data, reducing its ability to generalize. This behavior reflects the bias-variance trade-off, where increasing model flexibility reduces bias but significantly increases variance, leading to poor generalization.

VI. ERROR COMPARISON ACROSS ALL MODELS

Figure 10 shows the training and testing misclassification errors for all models tested in this experiment, plotted side by side for easy comparison.

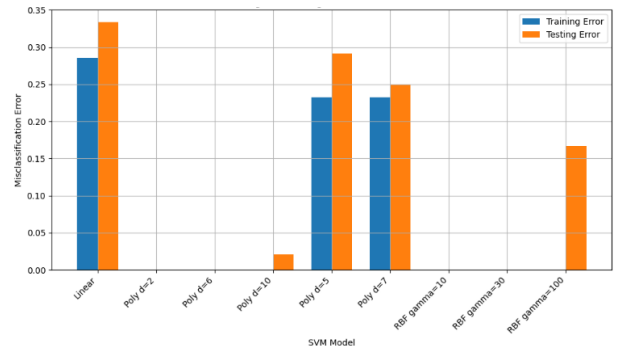


Fig. 10. Training and testing misclassification errors for all SVM models. Models with zero-height bars performed perfectly on both training and testing splits.

The bar chart makes the differences between models immediately clear. The linear kernel has the tallest bars, confirming that it is the worst performer overall. The odd-degree polynomial kernels (degrees 5 and 7) are the next worst group, with both training and testing errors around 23–29%. The even-degree polynomial kernel (degree 2) and the moderate RBF kernels ($\gamma = 10$ and 30) all have zero-height bars, indicating

perfect classification on both training and testing data. The only clear sign of overfitting appears in the RBF model with $\gamma = 100$, where the testing error rises to 16.67% while the training error remains at zero. This gap between training and testing error is a clear indicator of overfitting [2].

VII. DISCUSSION

Looking at the results overall, several clear patterns emerge.

The linear kernel fails completely, as expected. The data forms two concentric rings, and no straight line can separate them. This highlights that model choice must first match the structure of the data, not just rely on model capacity [4].

For the polynomial kernel, the key finding is the even-versus-odd behavior. The even-degree model (degree 2) produces a smooth circular boundary and achieves perfect classification, while odd degrees (5 and 7) generate distorted boundaries and perform poorly. This is not simply underfitting, but a structural mismatch: odd-degree terms break the rotational symmetry of the data, whereas even-degree terms preserve it. At degree 10, a small test error appears, indicating the onset of overfitting.

For the RBF kernel, the parameter γ controls how locally the model fits the data. At $\gamma = 10$ and 30, the model captures the overall structure and generalizes perfectly. However, at $\gamma = 100$, the decision boundary becomes overly localized, fitting individual training points and leading to a clear increase in testing error. This is a textbook example of overfitting [2].

Overall, these results show that effective kernel selection depends on aligning the model with the geometry of the data rather than increasing complexity.

VIII. CONCLUSION

In this experiment, we tested a range of SVM kernels on a circular dataset that cannot be separated with a straight line. We included a linear baseline, four polynomial kernel degrees, and three RBF gamma values, and compared their training and testing errors side by side.

The results were clear. The linear kernel failed entirely. Odd-degree polynomial kernels underfitted due to a structural symmetry mismatch with the data. The even-degree polynomial kernel (degree 2) and moderate-gamma RBF kernels achieved perfect classification. A higher-degree polynomial (degree 10) showed the onset of overfitting, while a high gamma in the RBF kernel led to clear overfitting, with testing error rising to 16.67% while training error remained at zero.

The main takeaway is that kernel selection should be guided by the structure of the problem, not just by model complexity. Understanding why a kernel works, not just whether it works, allows for better model selection in practice [3], [4].

Future work could explore automated model selection techniques to systematically identify optimal kernel parameters.

REFERENCES

- [1] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [2] V. Vapnik, *The Nature of Statistical Learning Theory*, 2nd ed. New York, NY: Springer, 1999.
- [3] B. Schölkopf and A. J. Smola, *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. Cambridge, MA: MIT Press, 2002.
- [4] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY: Springer, 2006.
- [5] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.